High-Temperature Sherrington–Kirkpatrick Cutoff

Physical Ram Moment Coordinates, Complementary-Space Spectral Certificates, and Cutoff-Rate Error Accounting

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Abstract

The source manuscript claims worst-start total-variation cutoff throughout the fixed high-temperature Sherrington–Kirkpatrick phase, with a rate defined by two dimension-first equilibrium spectral constructions. We examine its normalization, weighted exact-gradient space, passage to an ordinary gradient norm, isolation of slow modes, and final cutoff argument. Ram exponential moments give exact finite Gibbs and gradient coordinates; their physical Gram matrices cannot be discarded. We derive complementary- space lower bounds for a self-adjoint rate, a bounded-semigroup alternative, relative matrix-error bounds, one-sided spin-correlation information, explicit convolution estimates, and accumulated block-error bounds. These are conditional certificates with complete local proofs, not a new proof of the manuscript's high-dimensional probabilistic inputs. In particular, we do not numerically determine the cutoff constant, exclude all omitted slow modes, extend the result to the critical temperature, or establish a computational speed advantage. The distinction between applicability and realized contribution is maintained throughout. Contribution assessment: C2/5, a useful source-specific reformulation and error-accounting layer without a certified improvement to the original asymptotic cutoff theorem.

Status. This is a confidential technical study of manuscript M430 in the project's pinned OpenAI collection [1]. Applicability is L4; contribution is C2, on a different scale. Source claims, imported dependencies, proved local results, and finite diagnostics are distinguished. Classical spectral, variational, and matrix arguments are credited as such, not presented as new inventions merely because their coordinates are Ram moments.

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1 Problem, Quantifiers, and the Two Clocks

1.1 Finite model and physical space

Fix $0 \le \beta < 1$, independently of n. On $\Omega_n = \{-1,1\}^n$ take a symmetric matrix J with zero diagonal and independent upper-triangular entries $J_{ij} \sim N(0, \beta^2/n)$. The zero-field Gibbs measure and local conditional quantities are

$$\mu(x) = Z^{-1} \exp(\tfrac{1}{2} x^T J x), \quad m_i(x) = \tanh((Jx)_i), \quad v_i = 1 - m_i^2, \quad w_i = 1 - x_i m_i. \tag{1.1}$$

Z is a sum over configurations, not an expectation under uniform probability. This distinction changes the pressure by $\log 2$ but not the normalized measure. At finite n, every configuration has positive mass and $0 < v_i \le 1$; no dimension-uniform lower bound on v_i follows.

For functions on the cube, put

$$d_i f(x) = \tfrac{1}{2}\{f(x^{i,+}) - f(x^{i,-})\}, \qquad Lf = \sum_i (m_i - x_i) d_i f = \sum_i \tfrac{w_i}{2}(f(x^i) - f(x)).$$

Each site has a rate-one refresh clock. The state space of L is $L^2(\mu)$, with the whole finite-dimensional space as its domain. There are no spatial boundary conditions or gauge choices. The relevant constraint when discussing the positive scalar spectral gap is $\mu f = 0$: constants are the zero eigenspace and must be removed. The physical Dirichlet form is

$$\mathcal{E}(f,g) = -\langle f, Lg\rangle_\mu = \mathbb{E}_\mu \sum_i v_i \overline{d_i f}\, d_i g. \tag{1.2}$$

Inner products here are conjugate-linear in the first argument. Most source calculations are real; the complex convention matters in moment matrices and adjoints.

Lemma 1.1 (Conditional identities and normalization)**.** *For the model above,*

$$\mathbb{E}[w_i \mid x_{-i}] = \mathbb{E}[w_i^2 \mid x_{-i}] = v_i, \qquad d_i(fg) = f\, d_i g + g\, d_i f - 2x_i (d_i f)(d_i g).$$

Consequently (1.2) *holds. The single-site update-attempt kernel P satisfies $L = n(P - I)$, hence $S_t = e^{tL} = e^{nt(P-I)}$.*

Proof. Conditionally, x_i has mean m_i and square one. Expanding $(1 - x_i m_i)^2$ proves the first identities. Write $f = a + x_i b$, $g = c + x_i d$ with a, b, c, d independent of x_i; the derivative of their product is $ad + bc$, which is exactly the displayed product rule. Conditional covariance is $v_i \overline{d_i f} d_i g$, so summing conditional variances gives the form identity. A uniform choice of site refreshes one of the n rate-one summands with probability $1/n$, proving the clock relation. □

The replacement of w_i or w_i^2 by v_i requires stationarity and independence of the remaining integrand from spin i. It is not a pointwise identity and cannot be used under an arbitrary evolving law.

1.2 What the source claims

Let $d_n(t) = \max_x \|S_t(x,\cdot) - \mu\|_{\mathrm{TV}}$. The source's claim is that a deterministic $\lambda(\beta) > 0$ satisfies, for each fixed $\varepsilon \in (0,1)$,

$$d_n\left((1-\varepsilon)\frac{\log n}{2\lambda}\right) \longrightarrow 1, \qquad d_n\left((1+\varepsilon)\frac{\log n}{2\lambda}\right) \longrightarrow 0 \tag{1.3}$$

in probability over the disorder. In update attempts the location is $n \log n/(2\lambda)$. This is a worst-start statement on a common high-probability event, not only equilibrium relaxation or mixing for a typical randomly selected start. It specifies relative cutoff, not a sharp fixed-width window or a limiting profile.

The source constructs

$$\lambda = \min\{\lambda_{\mathrm{sp}}, \inf \sigma(A_v)\}, \qquad 0 < \lambda \le \lambda_{\mathrm{sp}} \le (1-\beta)^2. \tag{1.4}$$

Here λ_{sp} is a spin spectral edge and A_v is the generator with sign reversed on a completed exact-gradient space. Neither is, by definition, the smallest eigenvalue of a finite trial matrix. In particular, (1.4) does not assert $\lambda = (1-\beta)^2$. The finite scalar Gibbs gap is also not silently identified with this dimension-first cutoff rate.

1.3 An exact baseline at zero coupling

At $\beta = 0$, Walsh monomials $x_A = \prod_{i \in A} x_i$ obey $Lx_A = -|A|x_A$. Starting at a fixed configuration, the coordinates at time t are independent with means $x_i e^{-t}$. Direct multiplication of the likelihood ratios gives

$$1 + \chi^2(S_t(x,\cdot), \mu) = (1 + e^{-2t})^n. \tag{1.5}$$

At $t = (1+\varepsilon)\log n/2$ this tends to one, proving the upper cutoff via TV $\le \sqrt{\chi^2}/2$. At $t = (1-\varepsilon)\log n/2$, the test $n^{-1/2}\sum_i x_i X_i$ has mean $n^{\varepsilon/2}$ and variance at most one, whereas its stationary mean is zero and variance one. Chebyshev separation gives the lower cutoff. Thus $\lambda(0) = 1$. This checks the factor two and the factor n in the clocks without any high-temperature limiting input.

2 Source Proof Architecture and Its Irreducible Inputs

2.1 A dependency-aware reading

The target is the September 24 manuscript, archived at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Its nine main sections pass from static tilted laws to gradient paths, Gaussian calculus, row/spin limits, hidden gradients, exactness, and cutoff. The local report is not an independent certification of every companion theorem. In particular the all-high-temperature spectral-gap manuscript supplies an imported gap, and conditional/posterior gaps need their own stopped-localization estimates. A zero-field equilibrium gap does not automatically supply every posterior gap used in a localization proof.

Stage	Source role	Ram reformulation and remaining obligation
Static typicality	Quenched recipe limits, tilted laws, row and covariance controls	Moment coordinates encode the tested functions; common-event and entropy bounds still have to be proved.

Stage	Source role	Ram reformulation and remaining obligation
Gradient path	Exact half-difference evolution and rough vector estimates	Keep the discrete product correction and time-dependent physical law; ordinary differentiation rules would change the operator.
Spin/hidden split	A dimension-first spin spectral measure controls the hidden sector	Finite Laplace measurements have one-sided edge information; a quantitative lower edge is an extra certificate.
Exactness	Weighted potential gradients, no harmonic defect, smoothing	The weighted Gram is natural, but density of finite recipes is not a uniformly bounded inverse between the two completions.
Isolation	Sub-spin visible rates are isolated and lift to ordinary recipes	Complementary-space bounds must include every omitted low mode, not just the essential spectrum.
Cutoff	Uniform blocks, upper estimate, spin or prepared-visible lower test, clock transfer	Preserve the all-start quantifier, inverse-power floors, parameter order, and update-attempt normalization.

The independent source review read all 23 assigned TeX files, totaling 12,449 lines, plus the bibliography. It did not establish a concrete central error, but retained four substantial verification obligations: the companion's stable-field and posterior-gap inputs; the adaptive GOE diagram/norm control, particularly rare coordinates; the complete row/replica/network estimates underlying hidden-sector decay; and the unreplayed appendix interval computations. The appendix's alternative route is restricted to $\beta < 1/2$ and a cutoff ratio at its median scale. It is not an independent validation of the full-phase spectral rate. Reading all proofs and checking their stated dependencies is not equivalent to independently reconstructing every diagram count or executing every numerical certificate.

2.2 Two completions, not one norm

For vector recipes p, the source has ordinary and weighted scalar products

$$\|p\|_0^2 = \mathbb{E}_\mu \sum_i |p_i|^2, \qquad \|p\|_v^2 = \mathbb{E}_\mu \sum_i v_i |p_i|^2. \tag{2.1}$$

The limits define $\mathcal{H}$ and $\mathcal{H}_v$. The embedding $\iota : \mathcal{H} \to \mathcal{H}_v$ is continuous, injective, and dense in the source's construction; it is not asserted to have a bounded inverse. Its proof of injectivity uses a cutoff $v_i \geq \eta$ together with spatial tails for fixed recipes, then lets $\eta \downarrow 0$. It does not produce $\|p\|_0 \leq C\|p\|_v$ uniformly over all completed vectors.

The gradient evolution K_t satisfies $dS_t f = K_t df$. The exact core is the span of $K_t df_{B,\delta}$ for $t \geq 0$, $\delta > 0$, and smooth bounded scaled recipes B, using the source's test potentials. On its weighted closure $\mathcal{X}_v$, K_t is a positive self-adjoint semigroup, with $A_v = -\operatorname{gen}(K_t)$. Its domain is the strong-generator domain in $\mathcal{X}_v$:

$$D(A_v) = \{q : \lim_{t \downarrow 0} (q - K_t q)/t \text{ exists in } \mathcal{X}_v\}.$$

The form domain is $D(A_v^{1/2})$. These are not defined by arbitrary formal coefficient sequences. Positivity follows at the prelimit from $\langle df, K_t dg\rangle_v = \langle f, (-L)S_t g\rangle_\mu$, followed by the specified completion. The source's ordinary space $\mathcal{X}$ has additional analytic smoothing estimates; the weighted self-adjoint spectral theorem must not be transferred there by declaring the embedding unitary.

2.3 Why exactness is substantive

The source first proves saturation of weighted limits of stationary exact gradients. It then treats nonstationary vectors retaining a scalar coefficient Hilbert space: $q_s \in \mathcal{H} \otimes \mathcal{S}_s$, with transport $q_s = (K_{t-s} \otimes C_{s,t})q_t$. Replacing those random coefficients by deterministic ones loses information.

Good comparison times have both a gradient/Hessian budget and a reverse Hellinger cost, weighted by the vector magnitude. Adjoint curl tests and smoothing give a strong derivative closure. Weak convergence alone would not do so. Own-spin removal and a posterior no-harmonic-defect estimate control distance from a potential gradient by a weighted curl norm. At the independent endpoint the corresponding Hodge estimate has a factor $1/4$ with the ordered curl sum. Its transport uses two-spin masses and a stopped gap argument. A generic finite matrix inverse supplies none of those localization or curl estimates.

2.4 Isolation and the last cutoff step

The hidden-gradient estimate becomes an essential-norm bound only after weakly null sequences are converted to squared recipe-hidden outputs and smoothed. Dense intertwining, finite-dimensional Riesz pieces, and weighted self-adjointness identify the isolated rates below λ_{sp}. An essential spectral bound alone leaves open finitely many lower eigenvalues. This distinction motivates Section 4.

For the upper cutoff the source obtains a uniform fixed-block gradient contraction after fixed burn-in, with additive error n^{-H}. A normalization/contradiction argument keeps the polynomial envelopes and upgrades qualitative limits to this stated uniform conclusion; it is not a free quantitative consequence of convergence for a single recipe. At late times, field tails and an energy integral identify a time with controlled gradient norm. Propagation and Euclidean oscillation then make the final total variation small.

For the lower bound there are two cases. A positive amount of limiting spin spectral mass in a fixed interval just above its edge yields a separating linear observable. If a visible rate is slower, isolated eigenvectors are approximated by finite recipes and a short tilted preparation produces a ball of attainable means. The proof orders coarse recipes, block length, fine recipes, and preparation time carefully, so the Poincare constants do not become circular. A smallest-singular-value Jacobian bound gives the ball by lifting line segments with the inverse Jacobian ODE; global injectivity is unnecessary. Variance bounds then separate this prepared law from equilibrium. Convexity transfers that lower bound to the worst deterministic start.

Finally, Poissonization alone proves only one useful direction of clock transfer immediately. The source uses a uniform holding probability and a local shifted Poisson-to-binomial argument for the other. A coefficient representation does not justify dropping that part of the proof.

3 The Actual Ram Calculus Connection

3.1 Private passages used, and their scope

The applied corpus statements are the physical Gram and full-residual identities in Volume 21 [3], the spectral-limit and finite-to-global cautions in the same volume [4], and the concrete atom and implementation-margin results in TR248 [5]. These were read as mathematical statements with their proofs, not inferred from report titles. Corpus maps were used for routing, not as substitute proofs.

TR248 works with exponential-polynomial atoms, positive physical measures, finite model spaces, and quantified errors. In a nonorthogonal space the mass matrix is part of the problem, not a cosmetic preconditioner. Its finite-measurement obstruction explicitly prohibits inferring a continuum frame from finitely many positive Gram eigenvalues. Volume 21 likewise distinguishes a projected residual from a full residual, and strong resolvent convergence from uniform spectral exclusion. These are precisely the limitations relevant to $\mathcal{X}_v$ in the source.

The existing M428 study concerns critical autocorrelations at $\beta = 1$, with a different time scale and random limiting law. Its finite Gibbs normalization is compatible with the construction below, but its critical results are not used to prove a fixed-$\beta < 1$ cutoff, or conversely.

3.2 An exact finite moment realization

For finite n, define the entire exponential moment map

$$\mathcal{R}_\mu f(z) = \mathbb{E}_\mu[f(x)e^{z\cdot x}], \qquad z \in \mathbb{C}^n. \tag{3.1}$$

All derivatives may be moved inside the finite sum. If A is a subset of sites, $\partial_A \mathcal{R}_\mu f(0) = \mathbb{E}_\mu[f x_A]$. Thus Ram exponential moments contain the Gibbs-weighted Walsh measurements exactly. The multilinear monomials span all functions on the cube, since there are 2^n of them and the uniform Walsh matrix is invertible. Strict positivity of μ preserves linear independence in $L^2(\mu)$.

One may instead use restricted atoms $\psi_a(x) = w_a(x)e^{s_a\cdot x}p_a(x)$ and take moments of their products. Linear independence must be checked after restriction to the cube: $x_i^2 = 1$ makes many formal polynomials identical. An arbitrary list of more than 2^n atoms cannot be independent. There is no generic dimension reduction in the exact full realization.

For centered trial functions ψ_a, form

$$G_{ab} = \mathbb{E}_\mu \overline{\psi_a}\psi_b, \qquad K_{ab} = \mathbb{E}_\mu \sum_i v_i \overline{d_i\psi_a}\, d_i\psi_b. \tag{3.2}$$

Then $\dot{c} = -G^{-1}Kc$ is the conforming projected scalar evolution. For the full centered bank it is the exact finite chain. For a smaller bank it is a compression, not a closed equation for every initial state. The generalized eigenproblem is $Kc = \theta Gc$. Eigenvalues of K alone have no such physical interpretation.

For gradient recipes q_a, the corresponding two Gram matrices are

$$(G_v)_{ab} = \mathbb{E}_\mu \sum_i v_i \overline{q_{a,i}} q_{b,i}, \qquad (G_0)_{ab} = \mathbb{E}_\mu \sum_i \overline{q_{a,i}} q_{b,i}.$$

Both can be assembled from weighted moments in finite volume, but their limits need not be uniformly equivalent. A finite positive generalized eigenvalue of (G_v, G_0) does not make ι^{-1} bounded in the limit.

3.3 Physical synthesis and the full residual

Let A be self-adjoint on a Hilbert space H, and let $F : \mathbb{C}^m \to H$ be injective with columns in $D(A)$. Write

$$G = F^*F, \quad K = F^*AF, \quad B = G^{-1}K, \quad P = FG^{-1}F^*, \quad Q = I - P. \tag{3.3}$$

The letter F avoids confusing synthesis with the SK coupling J. Then $GB = B^*G$, and $G^{-1/2}KG^{-1/2}$ is Hermitian. For any real z and c,

$$\|(A - z)Fc\|^2 = \|(B - z)c\|_G^2 + \|QAFc\|^2. \tag{3.4}$$

Indeed the terms are the orthogonal P and Q projections of the full residual. This reproves the relevant Ram corpus identity in the present notation. In particular an exact generalized eigenpair need not be an eigenpair of A. Conversely, a small full residual locates *some* spectrum near z by the spectral theorem; it does not exclude spectrum strictly below that neighborhood.

4 Complementary-Space Certificates for a Slow Rate

4.1 A two-block lower bound with a proof

The following is a classical quadratic-form/Schur-complement estimate, organized for a finite Ram bank and its unrepresented complement. It applies in particular to A_v if its hypotheses can be certified there.

Theorem 4.1 (Global rate from a finite bank and a tail)**.** *Let A be a lower-semibounded self-adjoint operator with closed form a and form domain V. Let $S \subset D(A)$ be nonzero and finite dimensional, P its orthogonal projector, and $Q = I - P$. Suppose*

$$\begin{aligned} a(s, s) &\geq b\|s\|^2 && (s \in S), \\ a(q, q) &\geq \tau\|q\|^2 && (q \in V \cap S^\perp), \\ \|QAP\|_{S\to H} &\leq c. \end{aligned}$$

Set

$$\ell(b, \tau, c) = \frac{b + \tau - \sqrt{(b - \tau)^2 + 4c^2}}{2}. \tag{4.1}$$

Then $A \geq \ell I$ in form sense. If u is any upper bound for the smallest Rayleigh value on S, then $\ell \leq \inf \sigma(A) \leq u$.

Proof. Because $S \subset D(A) \subset V$, P and Q preserve V. For $f = s+q$, symmetry and the operator-form identity on S give

$$a(f, f) = a(s, s) + a(q, q) + 2\Re\langle q, As\rangle \geq b\|s\|^2 + \tau\|q\|^2 - 2c\|s\|\|q\|.$$

The smallest eigenvalue of $\left(\begin{smallmatrix} b & -c \\ -c & \tau \end{smallmatrix}\right)$ is (4.1), proving the lower bound for every $f \in V$. The form characterization of the spectral bottom proves the global claim. Restricting the infimum to S proves the upper bound. □

For a desired $z < \tau$, it is sufficient that

$$b - z - \frac{c^2}{\tau - z} > 0 \tag{4.2}$$

to prove $A > zI$ with a positive margin. This is useful with interval or rational bounds because it avoids an unnecessary square root. If the operator already has a known lower bound γ, replace ℓ by $\max(\gamma, \ell)$. A negative ℓ is still a valid lower bound, but would not certify a positive rate on its own.

For nonorthogonal synthesis, let $U = FG^{-1/2}$ be isometric. The finite diagonal block is $U^*AU = G^{-1/2}KG^{-1/2}$ and

$$c^2 \geq \|U^*A^2U - (U^*AU)^2\|. \tag{4.3}$$

Here U^*A^2U denotes the Gram matrix $(AU)^*(AU)$ and needs only $S \subset D(A)$, not $S \subset D(A^2)$. The difference equals $(QAU)^*(QAU)$, so its norm is exactly the squared cross norm. This notation convention is essential for an unbounded generator.

4.2 Two obstructions that the certificate cannot bypass

First, take $A = \operatorname{diag}(1, 2, 100)$ and $S = \operatorname{span}(e_2, e_3)$. Every retained eigenvector has zero full residual. The finite minimum is two, but the global minimum is one. The missing datum is $\tau = 1$; setting $\tau = 100$ from a supposed "fast tail" would be false.

Second, consider $Ae_1 = ae_1$ and $Ae_j = be_j$ for $j \geq 2$ on ℓ^2, with $0 < a < b$. The essential spectrum is $\{b\}$, yet a bank missing e_1 cannot use b as the complementary lower bound. Thus the source's essential spectral bound at the spin rate is not by itself a certificate for the whole complement of an arbitrarily selected Ram bank. All isolated modes below that edge must be covered or bounded separately.

These examples are positive, self-adjoint, and elementary. One need not invoke indefinite spectral pollution to demonstrate the present danger. The source's isolation theorem makes a finite-mode certificate plausible, but does not already provide its computable tail margin or a certified list of every low mode.

4.3 Relative matrix uncertainties

Proposition 4.2 (Mass-aware Rayleigh enclosure)**.** *Suppose $\widehat{G} > 0$ and Hermitian $\widehat{K}$ satisfy*

$$(1-\delta)\widehat{G} \preceq G \preceq (1+\delta)\widehat{G}, \qquad -\eta\widehat{G} \preceq K - \widehat{K} \preceq \eta\widehat{G}, \quad 0 \leq \delta < 1.$$

Let $\widehat{\theta}$ be the smallest eigenvalue of $\widehat{G}^{-1/2}\widehat{K}\widehat{G}^{-1/2}$ and assume $\widehat{\theta} \geq \eta \geq 0$. Then

$$\frac{\widehat{\theta} - \eta}{1+\delta} \leq \lambda_{\min}(G^{-1/2}KG^{-1/2}) \leq \frac{\widehat{\theta} + \eta}{1-\delta}. \tag{4.4}$$

Proof. Every numerator c^*Kc is at least $(\widehat{\theta} - \eta)c^*\widehat{G}c$, and the actual mass denominator is at most $(1+\delta)c^*\widehat{G}c$. Nonnegativity permits this division in the indicated direction. For the upper bound use a minimizing generalized eigenvector of the hatted pair, the upper numerator error, and the lower mass denominator. The Rayleigh infimum completes the proof. □

This proposition fixes a possible sign error in naive denominator bounds: if $\widehat{\theta} - \eta < 0$, its stated lower formula need not hold. One can instead use that negative numerator with the denominator $1-\delta$, or use an independent nonnegative form bound. Absolute entrywise errors become useful relative errors only after controlling $\widehat{G}^{-1/2}$. TR248's physical-margin principle supplies the appropriate computational discipline, not the missing SK tail theorem.

5 A Bounded-Semigroup Alternative

Working with $T_h = e^{-hA}$ can avoid estimating A on rough trial vectors. It cannot avoid controlling the complement.

Theorem 5.1 (Bounded-time spectral bracket)**.** *Let $A \geq 0$ be self-adjoint and $h > 0$. For a nonzero finite subspace $S \subset H$, let P, Q be its projections. Suppose*

$$\|PT_hP\| \leq b, \quad \|QT_hQ\| \leq \tau, \quad \|QT_hP\| \leq c, \quad 0 < l \leq \|PT_hP\|.$$

Put

$$U = \min\left\{1, \frac{b + \tau + \sqrt{(b-\tau)^2 + 4c^2}}{2}\right\}.$$

Then $l \leq \|T_h\| \leq U$ and

$$-h^{-1}\log U \leq \inf\sigma(A) \leq -h^{-1}\log l. \tag{5.1}$$

Proof. T_h is a positive self-adjoint contraction. Its quadratic form on $s+q$ is at most $b\|s\|^2 + \tau\|q\|^2 + 2c\|s\|\|q\|$, whose maximum on $\|s\|^2 + \|q\|^2 = 1$ is the displayed two-by-two upper eigenvalue. Also $\|T_h\| \leq 1$. Restriction gives the lower bound. The spectral theorem gives $\|e^{-hA}\| = e^{-h\inf\sigma(A)}$; logarithms yield the result. □

Unlike Theorem 4.1, no trial vector need belong to $D(A)$. However QT_hQ is not in general e^{-hQAQ}, since off-diagonal coupling affects the compressed dynamics. A complementary generator bound must be converted with that coupling accounted for, or the heat-block bound must be proved directly. Empirical agreement of two time-step resolutions is not such a proof. If $U = 1$, the result gives only the already known nonnegative lower bound; it does not certify a new gap.

In Ram coordinates, the finite heat matrix is $H_{ab}(h) = \langle\psi_a, K_h\psi_b\rangle_v$ and its physical representation is $G_v^{-1/2}H(h)G_v^{-1/2}$. Full heat residuals satisfy the analogue of (3.4). This provides a bounded-operator option for a recipe bank already available from the source's positive-time smoothing.

6 Spin Moments: What They Say About an Edge

The source's dimension-first stationary spin correlation has the form

$$C(t) = \int_{[0,\infty)} e^{-tu}\,\rho(du), \qquad \rho([0,\infty)) = 1, \quad a = \inf\operatorname{supp}\rho = \lambda_{\mathrm{sp}}. \tag{6.1}$$

The statements below apply to any probability measure on this half-line. They do not supply the model-specific convergence from C_n to C.

Proposition 6.1 (One-sided edge information)**.** *For $t \geq 0$ and $h > 0$,*

$$a \leq -\frac{1}{h}\log\frac{C(t+h)}{C(t)}.$$

Also $a \leq -\log C(t)/t$ for $t > 0$, and both expressions converge to a as $t \to \infty$ with h fixed in the first. The function $\log C(t)$ is convex on $(0,\infty)$.

Proof. The ratio is the Laplace transform at h of the tilted probability $e^{-tu}\rho(du)/C(t)$, supported on $[a,\infty)$, so it is at most e^{-ha}. The one-time bound follows in the same way. For every $\epsilon > 0$, $m = \rho([a, a+\epsilon/2]) > 0$. The tilted mass of $[a+\epsilon,\infty)$ is at most $m^{-1}e^{-t\epsilon/2}$, so the tilted law concentrates in $[a, a+\epsilon]$. Boundedness of e^{-hu} then proves the ratio limit. The inequalities $me^{-t(a+\epsilon/2)} \le C(t) \le e^{-ta}$ prove the one-time limit. Finally Holder's inequality for the defining integral proves log convexity; equivalently its second derivative is the variance of u under the tilt for $t > 0$, where exponential damping ensures finite moments. □

Proposition 6.2 (Finite-horizon invisibility of a slow component)**.** *For $0 \le a < b$ and $0 < \epsilon < 1$, set $\rho_0 = \delta_b$ and $\rho_\epsilon = (1-\epsilon)\delta_b + \epsilon\delta_a$. Their edges are b and a, but*

$$\sup_{t\ge 0} |C_\epsilon(t) - C_0(t)| \le \epsilon.$$

For $0 \le t \le T$ their relative discrepancy is at most $\epsilon(e^{(b-a)T} - 1)$.

Proof. Subtract the two explicit transforms: $C_\epsilon - C_0 = \epsilon(e^{-at} - e^{-bt})$. Divide by e^{-bt} for the relative assertion. Both bounds follow directly. □

Thus any finite absolute accuracy leaves room for an arbitrarily small slow spectral component. Exact knowledge of the whole transform is a different question. A numerical finite-time fit cannot convert Proposition 6.1's upper bound to a lower bound.

If one additionally has a certified edge-neighborhood mass $\rho([a, a+\delta]) \ge m > 0$, then

$$-\frac{\log C(t)}{t} - \delta + \frac{\log m}{t} \le a \le -\frac{\log C(t)}{t}. \tag{6.2}$$

The left bound is obtained from $C(t) \ge me^{-(a+\delta)t}$. This can be useful only when the mass statement is independently proved; it is not a way to infer that statement from the same data. If $C(t) \in [C_-, C_+]$ is a rigorous positive interval, use C_+ on the left of (6.2) and C_- on the right. This records measurement uncertainty without reversing monotonicity.

7 Ordinary-Norm Recovery and Uniform Block Errors

7.1 An explicit version of the unweighting convolution

The source recovers ordinary norm decay by a dissipative diagonal evolution and a bounded coupling from the weighted space, rather than by inverting ι. The following deterministic calculation makes its constants and the coincident-rate case visible.

Proposition 7.1 (Two-rate convolution)**.** *Suppose $a, b > 0$, $F_0, V_0, C \ge 0$, and a nonnegative function satisfies, for every $t \ge 0$,*

$$F(t) \le e^{-bt}F_0 + CV_0 \int_0^t e^{-b(t-s)}e^{-as}\,ds.$$

Then the integral is

$$I_{a,b}(t) = \begin{cases} (e^{-at} - e^{-bt})/(b-a), & a \ne b, \\ te^{-at}, & a = b. \end{cases} \tag{7.1}$$

For every $0 < \eta < \min(a,b)$,

$$F(t) \le \left(F_0 + \frac{CV_0}{e\eta}\right) e^{-(\min(a,b)-\eta)t}.$$

Proof. Integrate the scalar exponential, using its limit at $a = b$. Pointwise the integrand is at most $e^{-\min(a,b)t}$, hence $I_{a,b}(t) \le te^{-\min(a,b)t}$. Since $\sup_{t\ge 0} te^{-\eta t} = 1/(e\eta)$, the last bound follows. □

For M430, $b = 1 - \beta^2$ comes from diagonal dissipation; a is an admissible weighted decay rate. The source's edge bound $\lambda \le (1-\beta)^2 < 1 - \beta^2$ for $0 < \beta < 1$ explains why a strict slack below the relevant slow rate is available. A generic resonant estimate has a factor t and should not be rewritten as a constant times e^{-at} with no slack. At $\beta = 0$ the direct product chain already gives the exact answer without this detour.

7.2 Accumulated error in the source's block iteration

Proposition 7.2 (Uniform inexact backward blocks)**.** *Let $u_0, \dots, u_k$ be nonnegative numbers with*

$$u_j \le \alpha_j u_{j+1} + e_j, \qquad 0 \le \alpha_j \le \overline{\alpha} < 1, \quad 0 \le e_j \le E.$$

Then

$$u_0 \le \overline{\alpha}^k u_k + E\frac{1-\overline{\alpha}^k}{1-\overline{\alpha}}. \tag{7.2}$$

If a nominal contraction is α and a verified multiplicative error is at most ζ, it suffices that $\overline{\alpha} = \alpha + \zeta < 1$.

Proof. Substitute each inequality into the preceding one. The coefficient of e_j is the product of its preceding α_i, bounded by $\overline{\alpha}^j$. Sum the geometric series. □

Apply this with $u_j = N^2_{a+jD}$, nominal $\alpha = e^{-2(\lambda-\xi)D}$, and $E = n^{-H}$ only after the source's uniform hypotheses have been established. For $k = O(\log n)$, the error floor remains $O(n^{-H})$, with its explicit denominator. A compression residual checked under equilibrium is not a proof that the same E works for every evolving law, every restart, and every terminal test. That is the main logical restriction on a computational implementation of this otherwise elementary improvement in accounting.

For terminal norm $u_k \le n^r$, the sufficient target $u_0 \le n^{-q}$ is obtained, for example, from

$$\overline{\alpha}^k n^r \le \tfrac{1}{2}n^{-q}, \qquad E \le \tfrac{1}{2}(1-\overline{\alpha})n^{-q}.$$

This gives an explicit accuracy budget for any future validated solver. It is stronger bookkeeping than saying the local error is merely small, but it does not manufacture the missing uniform estimate.

8 From Spectral Intervals to a Cutoff Location

Theorem 8.1 (Conditional rate-to-time certificate)**.** *Suppose the source constructions and cutoff theorem hold, and rigorous bounds have been established for their* limiting *rates:*

$$0 < l_s \le \lambda_{\mathrm{sp}} \le u_s, \qquad 0 < l_v \le \inf \sigma(A_v) \le u_v.$$

With $l = \min(l_s, l_v)$ and $u = \min(u_s, u_v)$,

$$\frac{\log n}{2u} \le t_n = \frac{\log n}{2\lambda} \le \frac{\log n}{2l}. \tag{8.1}$$

If $|\widehat{\lambda} - \lambda| \le \delta < \widehat{\lambda}$, then

$$\frac{|\log n/(2\widehat{\lambda}) - t_n|}{t_n} \le \frac{\delta}{\widehat{\lambda}}.$$

The endpoints of the first interval are multiplied by n for update attempts. These statements bound the claimed cutoff location, not a finite-n mixing-time remainder or a cutoff window.

Proof. Taking a minimum preserves coordinatewise lower and upper bounds. The function $x \mapsto \log n/(2x)$ is decreasing for $x > 0$, proving (8.1). Direct cancellation gives the relative error $|\lambda - \widehat{\lambda}|/\widehat{\lambda}$. The clock factor was proved in Lemma 1.1. □

This theorem clarifies three distinct tasks: enclose the finite matrices, enclose the limiting operators and spin support, and transfer those enclosures into cutoff statements using the full source theorem. Completing the first is not completing the second or third.

8.1 A practical conditional algorithm

1. Fix the model, temperature, zero-field convention, clock, and source snapshot. Identify whether the desired target is a scalar finite gap, the limiting spin edge, or the limiting exact-gradient rate.
2. Select finite independent moment/recipe banks in the *correct* space. Assemble G_v, stiffness or heat matrices, and full residual quantities with rigorous quadrature and rounding bounds.
3. Certify a physical mass margin. Convert matrix errors using Proposition 4.2 or a comparable relative estimate, rather than substituting Euclidean coefficient norms.
4. Supply an analytic complementary-space lower bound for the generator, or an upper heat-block bound. Account explicitly for every possibly omitted isolated slow mode. Use Theorem 4.1 or Theorem 5.1 only after this step.
5. Separately certify spin-support information. A finite Laplace fit provides the upper bounds of Proposition 6.1, not the needed positive lower edge. If using (6.2), certify its mass input.
6. Apply Theorem 8.1. For approximate block propagation, retain the all-start norm and apply Proposition 7.2 with verified uniform errors.
7. Record all temperatures, dimensions, trial sizes, truncation parameters, disorder events, and numerical tolerances. Reject a claimed global certificate when any complementary or limit hypothesis is absent.

An implementation can stop successfully at a finite physical certificate without pretending the whole asymptotic program has been completed.

9 Extensions That Are Valid, and Extensions Still Open

9.1 General reversible finite systems

The moment identities and physical compression apply to any strictly positive finite Gibbs law with a reversible generator after substituting its actual Dirichlet form. They do not require Gaussian SK couplings. Likewise Theorems 4.1 and 5.1 are abstract self-adjoint facts. This is a genuine extension of the representation and certification layer, not a universal cutoff theorem for those models.

Adding an external field changes the physical Gibbs weights and the conditional means; spin-flip parity and the source's dimension-first construction may then change. Non-Gaussian disorder requires its own typicality and dynamic comparison estimates. A sequence $\beta_n \uparrow 1$ requires uniform constants in β, which the fixed-temperature argument does not supply. None of these extensions is obtained just by reusing the same matrix formulas.

9.2 Stability under a genuinely bounded perturbation

Proposition 9.1 (Same-space perturbation)**.** *Let A be self-adjoint and lower bounded on H, and let E be bounded self-adjoint with $\|E\| \le \epsilon$. Then, on the same domain,*

$$|\inf \sigma(A+E) - \inf \sigma(A)| \le \epsilon.$$

Proof. For each unit form vector, $|\langle f, Ef\rangle| \le \epsilon$. Take infima in both form inequalities. Bounded self-adjoint perturbation preserves self-adjointness on $D(A)$ and the form domain. □

The useful extension is an explicit robustness allowance for a proved same-space perturbation. Changing temperature changes measure, completion, and potentially the identified operators. One must construct a controlled identification first; this proposition is not a free Lipschitz theorem for $\lambda(\beta)$. Similar care is needed for changes of recipe banks: a coordinate change is not itself a physical operator perturbation.

9.3 Research targets with clear completion criteria

The highest-value next step is a complementary-space estimate that is computable for the exact-gradient generator. A proof giving explicit tail constants plus validated Gram/stiffness intervals would turn the conditional certificate into an actual enclosure of the cutoff rate. A second target is a quantitative lower bound on spin spectral mass near its edge, uniform over a fixed temperature interval. Together with (6.2), this would provide a finite-time edge error bound.

A third target is to quantify the source's qualitative normalizations and dimension thresholds sufficiently to allow β or trial size to vary with n. A fourth is a domain-correct reduced bank of isolated visible modes with a proved complement margin. None has been established here. Speed comparisons should be made against a whitened conventional Galerkin implementation of the same bank at the same physical accuracy; otherwise a coordinate change can be mistaken for a computational gain.

10 Finite Diagnostics and Reproducibility

The companion local script executed 650 assertions in eight groups, all passing. It checks the finite algebra of this report, not the high-dimensional SK theorem. The cases are deliberately small: exact rational Gram/residual examples, two-block bounds, the missed-ground-mode counterexample, a finite uncoupled heat-bath chain, relative matrix intervals, accumulated error recurrences, and the explicit two-atom spin-measure construction. Integer and rational checks are distinguished from floating-point evaluations of exponentials and spectra. The deterministic $n = 3$ Gibbs/form comparisons had maximum absolute discrepancy $3.06 \cdot 10^{-16}$ at tolerance 10^{-12}. The two-block floating tests used 20 positive definite five-dimensional fixtures and seed 430. The exact-arithmetic checks concerned rational identities, not interval evaluations of transcendental Gibbs weights. No interval-enclosed SK disorder simulation, limiting spectral support, or common-event concentration theorem is reported.

The analytic proofs, not enumeration of finitely many cases, establish the general statements above. The script and its result file are stored with the report's evidence. They make sign, clock, mass-matrix, and omitted-mode mistakes reproducible. A passed diagnostic is not a proof that the inputs needed for a real SK cutoff certificate exist.

11 Current Literature and Comparison

11.1 Primary records checked on October 8, 2026

The primary record for Wang [7], version 2 of August 28, states optimal-order random-site mixing for fixed $\beta < 1/2$, uniformly in external fields. Its described route uses an integrated Bakry–Emery criterion and a two-spin estimate, followed by a localization upgrade. This is not the same claim as zero-field cutoff with an identified dimension-first rate throughout $\beta < 1$.

Boban, Li, and Oveis Gharan [8], version 1 of September 11, describe a rank-one-perturbed trickledown method and polynomial mixing up to $1/2 + \varepsilon$ for a positive absolute ε. A polynomial bound is not an optimal cutoff location. Chen and Saloff-Coste [9] provide the earlier continuous/lazy-discrete cutoff-comparison context. These comparisons use the retrieved primary records and abstracts; they are not independent audits of those entire papers. The pinned target source is read more deeply and is still treated as a source claim with named dependencies, not as established solely by an announcement.

The project's October 8 upstream check found later repository changes, including withdrawals of three other catalogue manuscripts. The target M430 and its selected one-hop dependency files were not changed by that comparison. This file-level observation is not a certificate for the whole transitive mathematical dependency graph.

11.2 Advantages of the Ram formulation

- It expresses finite Gibbs and gradient measurements by one explicit exponential moment map, while retaining the actual measure and clock.
- It exposes both physical Gram matrices and the missing residual component, preventing a finite coefficient calculation from being misidentified with an infinite exact-gradient statement.
- It supplies compatible finite-block, tail, rounding, and time-location error formulas. The generator and bounded-semigroup options make their different domain requirements visible.
- It gives transparent negative controls: a zero full residual need not find the ground mode; essential spectrum does not bound every omitted mode; finite Laplace accuracy does not give a lower support edge.
- It makes assumptions that could be weakened explicit. A model-specific improvement of a tail or mass estimate can be inserted into the same certificate without changing the entire representation.

11.3 Costs and limitations relative to the source

The source's difficult achievements, if valid, are probabilistic and infinite-dimensional: posterior gaps, tilted typicality, exactness, hidden-sector control, isolated-mode lifting, and uniform worst-start propagation. The Ram coordinates do not replace them. The exact full finite basis is exponential in n; reduced banks need new tail proofs. Moment matrices may be ill-conditioned and their certification can cost more than a standard orthogonal-coordinate computation. No speedup, new temperature regime, narrower cutoff window, or new value of $\lambda(\beta)$ is obtained in this report.

The proved local inequalities are established mathematical tools applied in a source-specific way. They improve transparency and falsifiability, but that is not the same as a substantial extension of the original cutoff result. The C2 assessment reflects this distinction; L4 relevance does not force a high contribution score.

12 Conclusion and Audit Scope

There is a concrete Ram Calculus connection to the spectral and finite-representation stages of M430. The exact finite reformulation, complementary-space bounds, spin-edge limitations, and block-error accounting form a coherent conditional route to certified cutoff-rate computations. The decisive unresolved input is a quantitative global tail/support certificate for the limiting objects, not an additional finite matrix diagonalization. The original source's full cutoff claim has not been independently reproved here.

The parent reading covered the source wrapper and main Sections 1, 2, 8, and 9 in full, with an independent source-contract review covering the assigned build and bounded dependency checks. The private passages used directly are Volume 21's physical-metric and spectral-limit chapters in full, and TR248 lines 133–176, 349–419, and 1030–1143. Selected M428 introductory normalization and limitations were checked; the whole M428 proof is not newly claimed as read in this study. The evidence files record input hashes, detailed read scope, local theorem review, finite checks, and build results. Original sources were not altered, and private corpus passages were not transmitted to public literature services.

References

[1] OpenAI, *Cutoff throughout the high-temperature Sherrington–Kirkpatrick phase*, September 24, 2026. Pinned source: archived public build. Key source sections: `02-model-inputs.tex`, `08-exact.tex`, `09-cutoff.tex`; source labels `eq:exact-lambda`, `thm:cutoff-continuous`, `prop:cutoff-clock`.

[2] OpenAI, *A spectral gap throughout the high-temperature Sherrington–Kirkpatrick phase*, September 24, 2026. Companion cited by M430; its full theorem is an imported dependency. Pinned companion directory.

[3] M. Subbarao, Ram Calculus books, Volume 21, *A Nonorthogonal Representation Carries a Metric*, with subsequent physical residual and metric sections. Private local source `volumes/volume21/chapters/ch01d-physical-metrics.tex`.

[4] M. Subbarao, Ram Calculus books, Volume 21, *A Dense Sequence With a Spurious Spectral Value*, with subsequent strong-resolvent, domain, and dual-estimate sections. Private local source `volumes/volume21/chapters/ch01f-spectral-limits.tex`.

[5] M. Subbarao, *Stable Adaptive Ram Frames*, TR248. Private source `TR248-Stable-Adaptive-Ram-Frames.tex`; selected spaces, implementation margin, and moment-certificate passages identified in the audit scope. Not every result of that report is used here.

[6] M. Subbarao, *Functional Universality of Critical SK Autocorrelations: Physical Ram Coordinates, Dual-Energy Traces, and Domain-Aware Approximation Certificates*, TR-Ram-OpenAI-722-M428, 2026. Private project report; selected normalization and scope comparison only.

[7] S. Wang, *Optimal Mixing of Glauber Dynamics for the Sherrington–Kirkpatrick Model at* $\beta < 1/2$, arXiv:2608.22159v2, 2026. https://arxiv.org/abs/2608.22159v2.

[8] M. Boban, A. Li, S. Oveis Gharan, *Rank-1-perturbed trickledown theorems: Mixing time of Glauber dynamics for the Sherrington–Kirkpatrick model up to* $\beta \le 1/2 + \varepsilon$, arXiv:2609.13138v1, 2026. https://arxiv.org/abs/2609.13138v1.

[9] G.-Y. Chen, L. Saloff-Coste, *Comparison of cutoffs between lazy walks and Markovian semigroups*, arXiv:1304.4587, 2013. https://arxiv.org/abs/1304.4587.